Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Introduction

The viscous Burgers' equation

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad \nu > 0, \tag{1}$$

was introduced by Burgers [2] as a one-dimensional model of turbulence; it had appeared earlier in work of Bateman [1]. It has the form of the Navier–Stokes momentum equation without the pressure term, and combines nonlinear advection with viscous diffusion. Setting $\nu = 0$ gives the inviscid equation

$$u_t + u\,u_x = 0, \tag{2}$$

the scalar conservation law $u_t + f(u)_x = 0$ with flux $f(u) = u^2/2$. Conservation laws of this type arise in gas dynamics [5, 10] and traffic flow [11].

Classical solutions of (2) break down in finite time even for smooth data, after which only weak solutions make sense, and these are not unique. Physically, (2) idealises (1) for small ν: a shock is a layer of thickness $O(\nu)$, and the admissible weak solutions should be those that arise as limits of viscous solutions. We therefore address the following question: *why does the inviscid Burgers' equation require an entropy condition, and why is that the correct condition?* The answer given is that the entropy condition is exactly the criterion selected by vanishing viscosity. Solving (2) by characteristics (Section 2) gives the time at which classical solutions break down (Section 3). Beyond that time we work with weak solutions and the Rankine–Hugoniot condition (Section 4), whose non-uniqueness is exhibited in Section 5. The Lax and Oleinik entropy conditions of Section 6 restore uniqueness, and Section 7 uses them to solve the Riemann problem. Finally, Section 8 derives the Cole–Hopf formula, from which the entropy solution emerges as the vanishing viscosity limit (a result of Hopf, cited without proof), and examines Godunov's method numerically; the theory follows [5, §3.4] throughout.

2 Method of Characteristics

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{3}$$

with $u_0 \in C^1(\mathbb{R})$ in Sections 2 and 3. For discontinuous data, such as the Riemann data (19), (3) is understood in the weak sense of Definition 4.1.

2.1 Characteristics

A *characteristic* of a C^1 solution u is a curve $x(t)$ satisfying

$$\dot{x}(t) = u(x(t), t). \tag{4}$$

Along it,

$$\frac{d}{dt}\,u(x(t),t) = u_x\,\dot{x} + u_t = u_t + u\,u_x = 0. \tag{5}$$

Proposition 2.1 (Characteristics are straight lines)**.** *Let u be a C^1 solution of* (3) *on $\mathbb{R}\times[0,T)$, and let $x(t)$ solve* (4) *with $x(0)=x_0$. Then for $0\le t<T$,*

$$u(x(t),t)=u_0(x_0),\qquad x(t)=x_0+u_0(x_0)\,t. \tag{6}$$

Proof. By (5), $u(x(t),t)\equiv u_0(x_0)$, so (4) gives $\dot{x}(t)=u_0(x_0)$, which integrates to the stated formula. □

2.2 The implicit solution formula

Theorem 2.2 (Implicit solution via characteristics)**.** *Let $u_0\in C^1(\mathbb{R})$, $\Phi_t(x_0)=x_0+u_0(x_0)\,t$, and $T\in(0,\infty]$ such that for every $t\in[0,T)$, $\Phi_t'>0$ on $\mathbb{R}$ and Φ_t maps $\mathbb{R}$ onto $\mathbb{R}$. Then the function u defined implicitly by*

$$u(x,t)=u_0(x_0),\qquad \text{where } x=x_0+u_0(x_0)\,t, \tag{7}$$

is the unique C^1 solution of (3) *on $\mathbb{R}\times[0,T)$.*

Proof. Well-definedness and regularity. Since $\Phi_t'>0$, Φ_t is strictly increasing, hence bijective by hypothesis, so $u(x,t)=u_0(\Phi_t^{-1}(x))$ is well defined. The function $F(x_0,x,t)=x_0+u_0(x_0)t-x$ is C^1 with $\partial_{x_0}F=\Phi_t'>0$, so by the implicit function theorem $x_0(x,t)=\Phi_t^{-1}(x)$ is C^1 in (x,t) jointly (it is locally the unique root, and by injectivity coincides with the global one). Hence $u=u_0(x_0(x,t))$ is C^1.

The PDE holds. Write $x_0=x_0(x,t)=\Phi_t^{-1}(x)$. Differentiating $x=x_0+u_0(x_0)\,t$ in x and in t gives

$$\frac{\partial x_0}{\partial x}=\frac{1}{1+u_0'(x_0)\,t},\qquad \frac{\partial x_0}{\partial t}=\frac{-u_0(x_0)}{1+u_0'(x_0)\,t}, \tag{8}$$

both well defined since $\Phi_t'(x_0)=1+u_0'(x_0)t>0$. By the chain rule applied to $u(x,t)=u_0(x_0(x,t))$,

$$u_x=\frac{u_0'(x_0)}{1+u_0'(x_0)\,t},\qquad u_t=\frac{-u_0(x_0)\,u_0'(x_0)}{1+u_0'(x_0)\,t}, \tag{9}$$

so $u_t+u\,u_x=0$, and $u(x,0)=u_0(x)$ since $\Phi_0=\mathrm{id}$.

Uniqueness. Let v be any C^1 solution of (3) on $\mathbb{R}\times[0,T)$, and fix $t\in(0,T)$, $x\in\mathbb{R}$. Solve backwards in time the ODE $\dot\gamma(s)=v(\gamma(s),s)$, $\gamma(t)=x$ (the right-hand side is C^1, hence locally Lipschitz, so there is a unique maximal solution). By (5), v is constant along γ, hence so is $\dot\gamma$: γ is a straight line with bounded slope $v(x,t)$. It therefore stays bounded and cannot blow up, so it extends back to $s=0$. Setting $x_0=\gamma(0)$ we get $v(x,t)=v(x_0,0)=u_0(x_0)$ and $x=x_0+u_0(x_0)t$. Thus every point (x,t) lies on a characteristic coming from $t=0$, with $x_0=\Phi_t^{-1}(x)$, and $v(x,t)=u_0(\Phi_t^{-1}(x))=u(x,t)$. □

Corollary 2.3 (Local existence)**.** *Let $u_0\in C^1(\mathbb{R})$ with $m:=\inf_{x_0}u_0'(x_0)>-\infty$, and set $T=\infty$ if $m\ge0$ and $T=-1/m$ if $m<0$. Then* (3) *has a unique C^1 solution on $\mathbb{R}\times[0,T)$, given by* (7)*.*

Proof. Fix $t\in[0,T)$ and put $\kappa_t:=1+\min(m,0)\,t$, so $\kappa_t>0$ and $\Phi_t'(x_0)=1+u_0'(x_0)t\ge\kappa_t$ for every x_0. Two consequences follow. (i) Φ_t is strictly increasing, hence injective. (ii) By the mean value theorem $\Phi_t(x_0)-\Phi_t(0)\ge\kappa_t x_0$ for $x_0\ge0$ and $\le\kappa_t x_0$ for $x_0\le0$, so $\Phi_t(x_0)\to\pm\infty$ as $x_0\to\pm\infty$; by continuity Φ_t is onto $\mathbb{R}$. Hence the hypotheses of Theorem 2.2 hold on $[0,T)$. □

Remark 2.4. The uniform lower bound $\Phi_t'\ge\kappa_t>0$, not strict monotonicity alone, gives surjectivity: for $u_0(x)=(\arctan x-x)/t_0$ one has $\Phi_{t_0}(x)=\arctan x$, strictly increasing with $\Phi_{t_0}'>0$ but not onto $\mathbb{R}$. The hypothesis $m>-\infty$ cannot be dropped either: for $u_0(x)=x^2$, $\Phi_t(x)=x+tx^2$ is not injective for any $t>0$, since $\Phi_t'(x)<0$ for $x<-1/(2t)$.

3 Breakdown of Classical Solutions

3.1 Blow-up of the derivative and the envelope

By the proof of Theorem 2.2,

$$u_x(x,t)=\frac{u_0'(x_0)}{1+u_0'(x_0)\,t},\qquad x=x_0+u_0(x_0)\,t. \tag{10}$$

The formula holds as long as the classical solution exists (Corollary 2.3, that is, for $t<-1/\inf u_0'$). If $u_0'(x_0)\ge0$ the denominator is at least 1, and u_x remains bounded along that characteristic. If $u_0'(x_0)<0$, the right-hand side, regarded

as a formal expression along the characteristic through x_0, has a vanishing denominator at $t = -1/u_0'(x_0)$; the classical solution itself ceases to exist no later than this time, and strictly earlier unless x_0 minimises u_0' (Proposition 3.1). The characteristics

$$x(x_0, t) = x_0 + u_0(x_0)\, t, \qquad x_0 \in \mathbb{R}, \tag{11}$$

have an envelope where $\partial x/\partial x_0 = 1 + u_0'(x_0)\, t = 0$, that is, where $\Phi_t'(x_0) = 0$; this concerns the family (11), not the solution beyond its first breakdown.

3.2 The breakdown time

Proposition 3.1 (Breakdown time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) \in (-\infty, 0)$. Then the classical solution exists and is C^1 on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{12}$$

and $\sup_x |u_x(\cdot, t)| \to \infty$ as $t \uparrow T^*$. If the infimum is attained at $x_0 = a$ (any minimiser), then moreover no C^1 solution exists on $\mathbb{R} \times [0, T^* + \delta)$ for any $\delta > 0$, and the classical solution degenerates at time T^* at the points

$$x^* = a + u_0(a)\, T^*, \qquad t^* = T^*. \tag{13}$$

Proof. Since $r \mapsto -1/r$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every x_0 with $u_0'(x_0) < 0$. So $\Phi_t'(x_0) > 0$ for all x_0 throughout $[0, T^*)$, and Corollary 2.3 gives a classical solution u there. Choosing x_n with $u_0'(x_n) \to m$, (10) gives $u_x \to m/(1 + mt)$ at the point with foot x_n as $n \to \infty$, so $\sup_x |u_x(\cdot, t)| \geq |m|/(1 - |m|t) \to \infty$ as $t \uparrow T^*$. Now let the infimum be attained at a, and suppose a C^1 solution v existed on $\mathbb{R} \times [0, T^* + \delta)$. By the uniqueness part of Theorem 2.2, $v = u$ on $[0, T^*)$. Along the characteristic $x(t) = a + u_0(a)t$, (10) gives $u_x = m/(1 + mt) \to -\infty$ as $t \uparrow T^*$, so v_x would be unbounded on the compact set $\{(x(t), t) : 0 \leq t \leq T^*\}$, contradicting its continuity. Substituting $x_0 = a$, $t = T^*$ into (11) gives x^*. □

If the infimum is not attained there is no first point of degeneracy, and no claim is made about extension beyond T^*.

Remark 3.2. Assume that u_0 is bounded, that the infimum m of u_0' is attained, and that $\{u_0' = m\}$ contains no interval. Then $\Phi_{T^*}' \geq 0$ vanishes only on a set with empty interior, so Φ_{T^*} is strictly increasing; it is onto $\mathbb{R}$ because u_0 is bounded, so $\Phi_{T^*}(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$ (this replaces the argument of Corollary 2.3, which degenerates at $t = T^*$). Thus exactly one characteristic reaches each point at time T^*, and u, bounded by $\sup |u_0|$ by (6), develops at (x^*, T^*) only a gradient catastrophe, $|u_x| \to \infty$. If instead $u_0' \equiv m$ on an interval, then Φ_{T^*} is constant on it and characteristics carrying different values of u_0 meet at one point at time T^*, so one expects a discontinuity to form at T^*; this is not proved here (it follows from the Lax–Oleinik formula, Remark 8.5).

3.3 Example: $u_0 = -\arctan x$

Here $u_0'(x_0) = -1/(1 + x_0^2) \in [-1, 0)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$ and $x^* = 0$: the singularity first appears at $(x^*, t^*) = (0, 1)$ (Figure 1). Here $\Phi_t'(x_0) = 1 - t/(1 + x_0^2)$ first vanishes at $t = 1$, at $x_0 = 0$. At $t = 1$ the map $\Phi_1(x_0) = x_0 - \arctan x_0$ is strictly increasing but its inverse is not C^1 at 0; for $t > 1$, $\Phi_t' < 0$ on an interval around $x_0 = 0$, so Φ_t is not injective and characteristics cross.

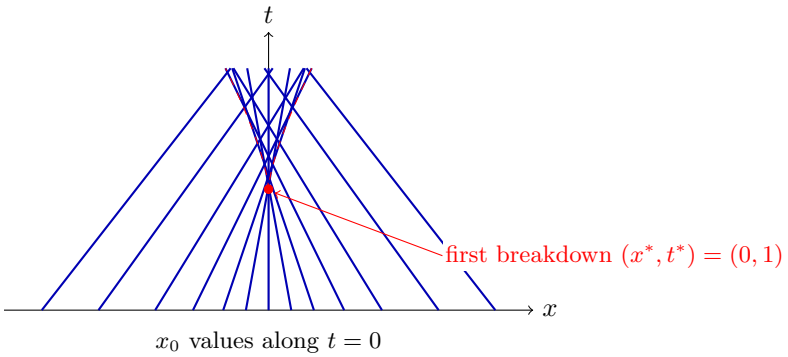


Figure 1: Characteristics $x = x_0 - \arctan(x_0)\, t$ for $u_0 = -\arctan x$, with their envelope (red, dashed), which has a cusp at the first breakdown $(x^*, t^*) = (0, 1)$. Inside the cusp ($t > 1$) three characteristics pass through each point.

By Remark 8.5 and Theorem 6.6, the entropy solution is given by the Lax–Oleinik formula (35). Here $V(y) = -\big(y \arctan y - \frac{1}{2}\ln(1+y^2)\big)$ is even, and

$$\partial_y G(x,y,t) = \frac{\Phi_t(y)-x}{t}, \qquad \partial_y^2 G = \frac{\Phi_t'(y)}{t} = \frac{1}{t} - \frac{1}{1+y^2}, \qquad \Phi_t(y) = y - t\arctan y.$$

$t \le 1$. Then $\partial_y^2 G > 0$ except possibly at $y = 0$, so $G(x,\cdot,t)$ is strictly convex and its unique minimiser is the unique critical point, the characteristic foot $\Phi_t^{-1}(x)$. Thus the entropy solution is the classical solution (7).

$t > 1$. Now Φ_t is odd, with $\Phi_t' < 0$ exactly on $(-s,s)$, $s = \sqrt{t-1}$. Fix $x > 0$. Since $G(x,y,t) - G(x,-y,t) = -2xy/t$, no minimiser has $y < 0$, and $y = 0$ is not one, since $\partial_y G(x,0,t) = -x/t < 0$. On $(0,s]$ we have $\Phi_t \le 0 < x$, and on (s,∞) Φ_t increases from a negative value to $+\infty$. Hence $G(x,\cdot,t)$ has exactly one critical point $y_+(x,t) > s$ in $(0,\infty)$, with $\Phi_t'(y_+) > 0$; it is the unique minimiser, and $u = (x-y_+)/t = -\arctan y_+$ is smooth in x by the implicit function theorem; by oddness the same holds for $x < 0$. At $x = 0$ the function $G(0,\cdot,t)$ is even with $\partial_y^2 G(0,0,t) = (1-t)/t < 0$, so 0 is not a minimiser, and the minimisers are $\pm y^*$, where $y^* > 0$ is the unique positive root of $y^* = t\arctan y^*$: a tie. Therefore, for $t > 1$, the entropy solution is smooth off $x = 0$ and has exactly one jump, stationary at $x = 0$ with

$$u_\ell = -u_r = \frac{y^*}{t} = \arctan y^* \in (0, \tfrac{\pi}{2}).$$

Moreover $y^* = \sqrt{3(1-1/t)}\,(1+o(1))$ as $t \downarrow 1$ from $\arctan y = y - y^3/3 + O(y^5)$, so $u_\ell \sim \sqrt{3(t-1)}$, and $y^* > s \to \infty$ gives $u_\ell \to \pi/2$ as $t \to \infty$, matching the limits $u_0(\mp\infty) = \pm\pi/2$.

4 Weak Solutions

4.1 The integral formulation

Let $f(u) = u^2/2$. If u is a C^1 solution of $u_t + f(u)_x = 0$ with $u(\cdot,0) = u_0$ and $\varphi \in C_c^1(\mathbb{R}\times[0,\infty))$, then integration by parts in x and t gives

$$0 = \int_0^\infty\!\!\int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi\,dx\,dt = -\int_0^\infty\!\!\int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\,dx\,dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\,dx. \tag{14}$$

Since the right-hand side requires no differentiability of u, it can be used to define solutions of lower regularity.

Definition 4.1 (Weak solution)**.** Let $u_0 \in L^\infty(\mathbb{R})$. A function $u \in L^\infty(\mathbb{R}\times(0,\infty))$ is a *weak solution* of (3) (with data u_0) if

$$\int_0^\infty\!\!\int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\,dx\,dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\,dx = 0 \tag{15}$$

for every $\varphi \in C_c^1(\mathbb{R}\times[0,\infty))$.

Reversing the integration by parts in (14) and applying the fundamental lemma of the calculus of variations shows that a bounded $u \in C^1(\mathbb{R}\times[0,\infty))$ is a weak solution with data u_0 if and only if it is a classical solution with $u(\cdot,0) = u_0$.

4.2 Piecewise smooth functions and the Rankine–Hugoniot condition

Definition 4.2 (Piecewise C^1 function)**.** Let $\sigma_1,\dots,\sigma_k : [0,\infty) \to \mathbb{R}$ be C^1 with $\sigma_1(t) < \cdots < \sigma_k(t)$ for $t > 0$ (the curves may start at a common point when $t = 0$). Put $\sigma_0 \equiv -\infty$, $\sigma_{k+1} \equiv +\infty$ and $\Omega_i = \{(x,t) :\ t > 0,\ \sigma_i(t) < x < \sigma_{i+1}(t)\}$ for $i = 0,\dots,k$. A bounded function u is *piecewise C^1 with respect to $\sigma_1,\dots,\sigma_k$* if on each Ω_i it is C^1 and u, u_x, u_t extend continuously to the closure of Ω_i in $\mathbb{R}\times(0,\infty)$, that is, to the points of the boundary curves with $t > 0$. Along $x = \sigma_i(t)$ we write $u_\ell(t)$ and $u_r(t)$ for the limits taken from Ω_{i-1} and Ω_i respectively.

Lemma 4.3 (Boundary identity)**.** *Let u be piecewise C^1 with respect to $\sigma_1,\dots,\sigma_k$ and satisfy $u_t + f(u)_x = 0$ classically in each Ω_i. Then for every $\varphi \in C_c^1(\mathbb{R}\times[0,\infty))$ and every $\varepsilon > 0$,*

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx + \sum_{i=1}^{k}\int_\varepsilon^\infty \Big[\big(f(u_\ell) - \sigma_i' u_\ell\big) - \big(f(u_r) - \sigma_i' u_r\big)\Big]\varphi(\sigma_i(t),t)\,dt. \tag{16}$$

Proof. On each $\Omega_i^\varepsilon = \Omega_i \cap \{t > \varepsilon\}$ (a Lipschitz domain, bounded after intersecting with $\operatorname{supp}\varphi$) the vector field $\varphi\,(f(u), u)$ (components in the (x,t) order) is C^1 up to the boundary, and its divergence is $u\varphi_t + f(u)\varphi_x$ because $u_t + f(u)_x = 0$ there. The divergence theorem gives

$$\iint_{\Omega_i^\varepsilon} (u\varphi_t + f(u)\varphi_x)\,dx\,dt = \oint \varphi\,\big(f(u)\,n_x + u\,n_t\big)\,d\sigma,$$

where $d\sigma\,\mathbf{n}$ equals $(1, -\sigma_{i+1}')\,dt$ on the right edge $x = \sigma_{i+1}(t)$, $(-1, \sigma_i')\,dt$ on the left edge $x = \sigma_i(t)$, and $(0,-1)\,dx$ on the bottom edge $t = \varepsilon$. Summing over i, the bottom edges give $-\int u(x,\varepsilon)\varphi(x,\varepsilon)\,dx$, and the two sides of each curve $x = \sigma_i(t)$ (from Ω_{i-1} on its left, Ω_i on its right) combine into the bracket in (16). □

Theorem 4.4 (Rankine–Hugoniot condition)**.** *Let u be piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$, and a weak solution of (3) for some data $u_0 \in L^\infty$. Then $u_t + f(u)_x = 0$ classically in each Ω_i, and along each curve $x = \sigma_i(t)$, $t > 0$,*

$$f(u_\ell(t)) - f(u_r(t)) = \sigma_i'(t)\,\big(u_\ell(t) - u_r(t)\big). \tag{17}$$

In particular, wherever $u_\ell(t) \neq u_r(t)$, and for $f(u) = u^2/2$,

$$\sigma_i'(t) = \frac{f(u_\ell) - f(u_r)}{u_\ell - u_r} = \frac{u_\ell(t) + u_r(t)}{2}. \tag{18}$$

Proof. Test functions φ supported in a single Ω_i give $\iint (u\varphi_t + f(u)\varphi_x) = 0$, so $u_t + f(u)_x = 0$ in the weak sense there; as u is C^1 on Ω_i, this holds classically, so Lemma 4.3 applies.

Now fix $t_0 > 0$, an index i, and $\varphi \in C_c^1(\mathbb{R} \times (0,\infty))$ supported in a small ball around $(\sigma_i(t_0), t_0)$ meeting no other curve. Take ε below the support of φ in (16): the left side is 0 by (15) (as $\varphi(\cdot,0) = 0$), the bottom term vanishes, and only the curve $x = \sigma_i(t)$ contributes to the sum, so

$$0 = \int_0^\infty \Big[\big(f(u_\ell) - \sigma_i' u_\ell\big) - \big(f(u_r) - \sigma_i' u_r\big)\Big]\varphi(\sigma_i(t), t)\,dt.$$

Since $\varphi(\sigma_i(t), t)$ may be any C^1 function of t with small support, the bracket vanishes identically, which is (17). The first form of (18) follows by division, and the second from $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$. □

Proposition 4.5 (Piecewise smooth weak solutions)**.** *Let $u_0 \in L^\infty(\mathbb{R})$ and let u be piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$. Suppose*

(i) *$u_t + f(u)_x = 0$ classically in each Ω_i;*

(ii) *the jump relation (17) holds along each curve $x = \sigma_i(t)$, $t > 0$ (automatic where u is continuous across the curve);*

(iii) *$u(x,t) \to u_0(x)$ as $t \to 0^+$ for every $x \notin \{\sigma_1(0), \dots, \sigma_k(0)\}$.*

Then u is a weak solution of (3) with data u_0.

Proof. Fix $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and $\varepsilon > 0$. By (i), Lemma 4.3 applies, and by (ii) every term of the sum in (16) vanishes, so

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx.$$

Let $\varepsilon \to 0^+$. The left side tends to the integral over $t > 0$ (bounded integrand, compact support). On the right, $\varphi(\cdot,\varepsilon) \to \varphi(\cdot,0)$ uniformly and, by (iii), $u(\cdot,\varepsilon) \to u_0$ almost everywhere with $|u| \le \|u\|_\infty$, so dominated convergence gives $-\int u_0\varphi(\cdot,0)\,dx$. This is (15); since the argument is run on $t > \varepsilon$, the curves are allowed to meet at $t = 0$. □

5 Non-Uniqueness of Weak Solutions

Consider (3) with the Riemann data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{19}$$

Characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so none covers the wedge $0 < x < t$.

Proposition 5.1. *The function*

$$u_A(x,t) = \begin{cases} 0, & x < t/2, \\ 1, & x > t/2, \end{cases} \tag{20}$$

is a weak solution of (3) *with data* (19).

Proof. Apply Proposition 4.5 with $k = 1$ and $\sigma_1(t) = t/2$: u_A is constant on each side, the jump moves at the Rankine–Hugoniot speed $(0+1)/2 = 1/2$ of (18), and $u_A(\cdot,t) \to u_0$ at every $x \neq 0$. □

Proposition 5.2. *The function*

$$u_B(x,t) = \begin{cases} 0, & x \leq 0, \\ x/t, & 0 < x < t, \\ 1, & x \geq t, \end{cases} \tag{21}$$

is a weak solution of (3) *with data* (19).

Proof. Apply Proposition 4.5 with $k = 2$, $\sigma_1 \equiv 0$, $\sigma_2(t) = t$: u_B is constant on the outer regions, solves the PDE on $0 < x < t$ since $(u_B)_t + u_B(u_B)_x = -x/t^2 + (x/t)(1/t) = 0$, is continuous across both curves (so (17) holds trivially), and $u_B(\cdot,t) \to u_0$ at every $x \neq 0$. □

Corollary 5.3 (Non-uniqueness)**.** *The initial value problem* (3) *with data* (19) *has at least two distinct weak solutions, u_A and u_B, which differ on the wedge $0 < x < t$.*

Remark 5.4 (An infinite family). For every $\alpha \in \mathbb{R}$ let

$$u^{(\alpha)}(x,t) = \begin{cases} 0, & x < \frac{\alpha}{2}t, \\ \alpha, & \frac{\alpha}{2}t < x < \frac{1+\alpha}{2}t, \\ 1, & x > \frac{1+\alpha}{2}t. \end{cases}$$

The jumps $0 \to \alpha$ and $\alpha \to 1$ move at their Rankine–Hugoniot speeds $\alpha/2 < (1+\alpha)/2$, so Proposition 4.5 (with $k = 2$) shows that each $u^{(\alpha)}$ is a weak solution with data (19). These are distinct for distinct α, except that $u^{(0)} = u^{(1)} = u_A$.

6 The Entropy Condition

6.1 The Lax shock condition

Definition 6.1 (Lax shock condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with $f \in C^2$, $f'' > 0$, and suppose u has a jump discontinuity along a C^1 curve $x = \sigma(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax shock condition* if

$$f'(u_\ell(t)) > \sigma'(t) > f'(u_r(t)) \tag{22}$$

for all t.

For strictly convex f the Rankine–Hugoniot speed $(f(u_\ell) - f(u_r))/(u_\ell - u_r)$ lies strictly between $f'(u_\ell)$ and $f'(u_r)$, in the order of u_ℓ, u_r (mean value theorem), so (22) holds if and only if $u_\ell > u_r$. For $f(u) = u^2/2$ and $\sigma' = (u_\ell + u_r)/2$, (22) reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{23}$$

Thus characteristics on both sides run into the shock. Since the Lax condition constrains jumps only, its derivation from viscosity, giving $u_\ell > u_r$, is deferred to Proposition 8.7.

6.2 Entropy solutions: the Oleinik condition

Definition 6.2 (Entropy solution)**.** A weak solution $u \in L^\infty$ of (3) is an *entropy solution* if

$$u(x+z,t) - u(x,t) \;\leq\; \frac{z}{t} \qquad \text{for every } z > 0 \text{ and a.e. } (x,t) \in \mathbb{R} \times (0,\infty). \tag{24}$$

The one-sided Lipschitz bound (24) is due to Oleinik [12]. It prevents $u(\cdot,t)$ from increasing at a rate exceeding $1/t$, which is what excludes upward jumps (Lemma 6.3).

Lemma 6.3 (Oleinik implies Lax)**.** *Let u be a piecewise C^1 entropy solution of Burgers' equation. Then every jump of u satisfies the Lax condition* (22)*, equivalently $u_\ell > u_r$.*

Proof. Fix a curve $x = \sigma(t)$ and $t_0 > 0$ with $u_\ell(t_0) \neq u_r(t_0)$. For $z > 0$ small enough that $[\sigma(t_0) - z, \sigma(t_0) + z]$ meets no other curve, there is a neighbourhood N of $(\sigma(t_0), t_0)$ in which every pair $(x,t), (x+z,t)$ with $x < \sigma(t) < x + z$ lies on opposite sides of the curve $x = \sigma(t)$ only, where u is C^1. Both sides of (24) are continuous on the open set $\{(x,t) \in N : x < \sigma(t) < x + z\}$, so the a.e. inequality holds at every point of it, in particular at $(\sigma(t_0) - z/2, t_0)$. Letting $z \to 0$, the left side tends to $u_r(t_0) - u_\ell(t_0)$ and the right side to 0. Hence $u_r \leq u_\ell$ at t_0, and since the limits differ, $u_\ell > u_r$. □

Remark 6.4. For a piecewise C^1 solution with locally finitely many jumps, (24) holds if and only if each jump satisfies $u_\ell > u_r$ and $u_x \leq 1/t$ on the smooth pieces. The Lax condition is thus a local admissibility condition, pointwise on each jump, and is the local form of (24) at jumps for piecewise C^1 solutions. The Oleinik bound is global in x: it also constrains the smooth parts ($u_x \leq 1/t$), and it is meaningful for general L^∞ weak solutions, for which no jump structure is available. It is the bound used in the uniqueness proof below.

Proposition 6.5. *For the Riemann data* (19) *($u_\ell = 0 < u_r = 1$), the rarefaction solution u_B of* (21) *is an entropy solution, while the shock solution u_A of* (20) *and every $u^{(\alpha)}$ of Remark 5.4 are not.*

Proof. u_B is entropy. Write $u_B(x,t) = \min\{\max\{x/t, 0\}, 1\}$, the composition of $x \mapsto x/t$ with the 1-Lipschitz function $y \mapsto \min\{\max\{y, 0\}, 1\}$. Hence $u_B(\cdot, t)$ is $(1/t)$-Lipschitz and $u_B(x+z,t) - u_B(x,t) \leq z/t$, which is (24).

u_A is not. The jump $0 \to 1$ has $u_\ell < u_r$, violating (23), so by Lemma 6.3 u_A cannot be an entropy solution. Directly: for $z < t$ and every $x \in (t/2 - z, t/2)$, a set of positive measure, $u_A(x+z,t) - u_A(x,t) = 1 > z/t$.

$u^{(\alpha)}$ is not. Each $u^{(\alpha)}$ contains an upward jump: if $\alpha > 0$ the jump $0 \to \alpha$ is upward, and if $\alpha \leq 0$ the jump $\alpha \to 1$ is, so the argument used for u_A applies. □

6.3 Uniqueness

Theorem 6.6 (Oleinik; [5, §3.4.3])**.** *For $u_0 \in L^\infty(\mathbb{R})$ and $f(u) = u^2/2$, the initial value problem $u_t + f(u)_x = 0$, $u(\cdot, 0) = u_0$, has at most one entropy solution in the sense of Definition 6.2, up to a set of measure zero. An entropy solution exists: the Lax–Oleinik solution [5, §3.4.2] is a weak solution satisfying* (24) *(Remark 8.5).*

For a general uniformly convex flux, Oleinik's bound reads $u(x+z,t) - u(x,t) \leq C(1 + 1/t)\, z$, and uniqueness is proved in [5, §3.4.3] in this larger class, which contains every solution of (24) (take $C = 1$); this gives the uniqueness statement above.

Outline of the proof [5, §3.4.3]. Let u_1, u_2 be entropy solutions with the same data and $w = u_1 - u_2$. Since $f(u_1) - f(u_2) = bw$ with $b = \frac{1}{2}(u_1 + u_2)$, subtracting the weak formulations gives $\iint w\,(\varphi_t + b\,\varphi_x)\,dx\,dt = 0$ for all test functions φ. Given $\psi \in C_c^\infty(\mathbb{R} \times (0,T))$, one solves the dual transport problem $v_t + b_\varepsilon v_x = \psi$, $v(\cdot, T) = 0$, with b_ε a mollification of b. The Oleinik bound gives $b_{\varepsilon,x} \leq C/t$, which controls the transport backward in time: for each $s > 0$, $\int |v_x^\varepsilon(x,t)|\,dx \leq C_s$ for $s \leq t \leq T$, uniformly in ε. This bound lets the error $\iint w(b_\varepsilon - b)v_x^\varepsilon$ vanish as $\varepsilon \to 0$ and the contribution of $t < s$ be made small, so $\iint w\psi = 0$ for all ψ and $w = 0$ a.e. Without the one-sided bound the dual solution can develop unbounded variation, and the argument fails, as it must by Section 5. Kružkov's formulation via entropy inequalities [8] instead yields an L^1-contraction, $\|u_1(t) - u_2(t)\|_{L^1} \leq \|u_1(0) - u_2(0)\|_{L^1}$, and covers non-convex fluxes.

7 The Riemann Problem

The entropy solution is now determined for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, that is, for the data

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{25}$$

7.1 The rarefaction fan

Seek self-similar solutions $u(x,t) = g(\xi)$, $\xi = x/t$. Then

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{26}$$

so on each interval of ξ either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$), which, being increasing in x, can only connect states with $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (3) *with data* (25) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

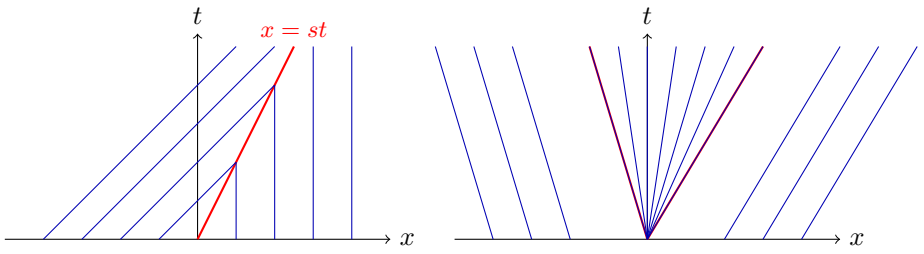
$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{27}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{28}$$

Proof. (a) Apply Proposition 4.5 with $k = 1$, $\sigma_1(t) = st$: u is constant on each side (so (i) holds), the jump moves at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ (so (ii) holds), and for $x < 0$ (resp. $x > 0$) we have $x < st$ (resp. $x > st$) for small t, giving (iii), so that u is a weak solution. It is moreover an entropy solution, since the only possible increments of $u(\cdot, t)$ are 0 and $u_r - u_\ell < 0$, whence $u(x + z, t) - u(x, t) \le 0 \le z/t$. By Theorem 6.6 it is the only entropy solution; its characteristics are shown in Figure 2(a).

(b) The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. Apply Proposition 4.5 with $k = 2$, $\sigma_1(t) = u_\ell t$, $\sigma_2(t) = u_r t$ (starting at a common point). (i) u is constant on the outer regions and equals x/t on the fan $u_\ell t < x < u_r t$, where it solves the PDE by Section 7.1. (ii) u is continuous across both curves, so (17) holds trivially. (iii) For fixed $x < 0$ we have $x < u_\ell t$ for all small t, and for fixed $x > 0$, $x > u_r t$ for all small t, so $u(\cdot, t) \to u_0$ away from 0. Hence u is a weak solution. It is an entropy solution because $u(x,t) = \min\{\max\{x/t, u_\ell\}, u_r\}$ is $(1/t)$-Lipschitz in x, so (24) holds, and uniqueness again follows from Theorem 6.6 (see Figure 2(b)). □



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristics for the two cases of Theorem 7.1. (a) $u_\ell = 1$, $u_r = 0$, $s = 1/2$: characteristics on both sides run into the shock $x = st$. (b) Characteristics fan out from the origin.

8 Extensions

This section tests the entropy condition against the two standard selection mechanisms: vanishing viscosity (Subsections 8.1–8.3) and monotone numerical approximation (Subsection 8.4).

8.1 The Cole–Hopf transformation

Theorem 8.1 (Cole–Hopf transformation [3, 7])**.** *Let $u \in C^{2,1}(\mathbb{R} \times (0,T))$ solve $u_t + u\,u_x = \nu\, u_{xx}$, with u and u_x continuous on $\mathbb{R} \times [0,T)$ (so that c below is defined), and set*

$$\hat\psi(x,t) = -\frac{1}{2\nu}\int_0^x u(y,t)\,dy, \qquad c(t) = \int_0^t \Big(\frac{u(0,\tau)^2}{4\nu} - \frac{u_x(0,\tau)}{2}\Big)d\tau, \qquad \theta(x,t) = \exp\big(\hat\psi(x,t) + c(t)\big) > 0. \tag{29}$$

Then θ solves the heat equation $\theta_t = \nu\,\theta_{xx}$, and $u = -2\nu\,\theta_x/\theta$.

Proof. Since c depends on t only, $\theta_x/\theta = \hat\psi_x = -u/(2\nu)$, which gives $u = -2\nu\theta_x/\theta$. For the heat equation, let $w = \hat\psi_t - \nu\hat\psi_{xx} - \nu\hat\psi_x^2$. Using $\hat\psi_x = -u/(2\nu)$, $\hat\psi_{xx} = -u_x/(2\nu)$, $\hat\psi_{xt} = -u_t/(2\nu)$ and $\hat\psi_{xxx} = -u_{xx}/(2\nu)$,

$$w_x = \hat\psi_{xt} - \nu\hat\psi_{xxx} - 2\nu\hat\psi_x\hat\psi_{xx} = -\frac{1}{2\nu}\big(u_t + u\,u_x - \nu\,u_{xx}\big) = 0. \tag{30}$$

Hence w depends on t only and may be evaluated at $x = 0$, where $\hat\psi(0,t) = 0$ gives $\hat\psi_t(0,t) = 0$:

$$w(t) = w(0,t) = \frac{u_x(0,t)}{2} - \frac{u(0,t)^2}{4\nu} =: C(t). \tag{31}$$

With $\psi = \hat\psi + c$ we have $c' = -C$, so $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = C + c' = 0$. Since $\theta = e^{\psi}$ satisfies $\theta_t = \theta\psi_t$ and $\theta_{xx} = \theta(\psi_{xx} + \psi_x^2)$, this is $\theta_t = \nu\theta_{xx}$. □

Remark 8.2. The factor $e^{c(t)}$ is independent of x, so it affects neither $u = -2\nu\theta_x/\theta$ nor the initial data ($c(0) = 0$). It cannot be omitted: $e^{\hat\psi}$ satisfies $\partial_t\theta = \nu\theta_{xx} + C(t)\theta$, which is the heat equation only if $C \equiv 0$.

8.2 Exact solution via the heat kernel

By Theorem 8.1, a classical solution of (1) that is bounded on $\mathbb{R}\times[0,T']$ for each $T' < T$ (this is what makes θ obey the growth bound below) corresponds to a solution of the heat equation with initial data

$$\theta_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right) \le e^{\|u_0\|_\infty |x|/(2\nu)}. \tag{32}$$

The Cauchy problem for the heat equation has a unique solution in the class of at most $e^{a|x|^2}$ growth [5, §2.3], which contains this data; it is given by

$$\theta(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\theta_0(y)\,dy. \tag{33}$$

Substituting (32)–(33) into $u = -2\nu\,\theta_x/\theta$ and differentiating under the integral gives the *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(r)\,dr + \frac{(x-y)^2}{2t}. \tag{34}$$

Corollary 8.3 (Global smoothness; Hopf [7])**.** *For every bounded measurable u_0 and every $\nu > 0$, formula* (34) *defines a C^∞ solution $u^{(\nu)}$ of* (1) *on $\mathbb{R}\times(0,\infty)$. Moreover $\|u^{(\nu)}\|_\infty \le \|u_0\|_\infty$, $u^{(\nu)}(\cdot,t) \to u_0$ in L^1_{loc} and at every continuity point of u_0 as $t \to 0^+$, and $u^{(\nu)}$ is the unique bounded solution with these properties.*

For $t > 0$, θ in (33) is smooth and positive, and for any such solution θ of the heat equation $u = -2\nu\theta_x/\theta$ solves (1): with $\psi = \ln\theta$, $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = 0$, and differentiating in x gives $(-u/2\nu)_t - \nu(-u/2\nu)_{xx} - \nu\big((u/2\nu)^2\big)_x = 0$, i.e. $u_t + u\,u_x = \nu u_{xx}$. This proves the first assertion for merely measurable u_0 (Theorem 8.1 is not needed in this direction), while the bound, the sense in which the data are attained, and uniqueness are taken from Hopf [7].

8.3 The vanishing viscosity limit

Theorem 8.4 (Vanishing viscosity limit [7])**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let $u^{(\nu)}$ be the solution of* (1) *with data u_0 given by* (34)*, and let u be the entropy solution of* (2) *with the same data (Definition 6.2; for Riemann data, Theorem 7.1). Then $u^{(\nu)} \to u$ pointwise almost everywhere and in L^1_{loc} as $\nu \to 0^+$.*

Hopf's proof [7] applies Laplace's method to (34) and shows that $u^{(\nu)}$ converges to the Lax–Oleinik solution (35). Because his work predates Oleinik's condition, the identification of this limit with u combines the theorem with Remark 8.5 below, which shows that the Lax–Oleinik solution satisfies (24), and with Theorem 6.6.

Remark 8.5 (The Lax–Oleinik formula). As $\nu \to 0$ the weight $\exp(-G/2\nu)$ in (34) concentrates at the minimisers of $y \mapsto G(x,y,t)$. Where the minimiser $y^*(x,t)$ is unique, the limit is

$$u(x,t) = \frac{x - y^*(x,t)}{t}, \qquad y^*(x,t) = \arg\min_y \left\{V(y) + \frac{(x-y)^2}{2t}\right\}, \quad V(y) = \int_0^y u_0, \tag{35}$$

the *Lax–Oleinik formula* [9, 12], [5, §3.4.2]; a jump occurs where two minimisers tie. Since $x \mapsto y^*(x,t)$ is nondecreasing [5, §3.4.2], for $z > 0$

$$u(x+z,t) - u(x,t) = \frac{z - \big(y^*(x+z,t) - y^*(x,t)\big)}{t} \;\le\; \frac{z}{t},$$

which is (24). As (35) also defines a weak solution [5, §3.4.2], the Lax–Oleinik solution is an entropy solution in the sense of Definition 6.2. The same bound shows $u(x+,t) - u(x-,t) \le 0$ wherever one-sided limits exist: every jump is downward. The minimiser is unique for almost every (x,t) [5, §3.4.2], so (35) defines u a.e.

Remark 8.6 (The Riemann problem from the Lax–Oleinik formula). For the data (25), $V(y) = u_\ell y$ for $y < 0$ and $V(y) = u_r y$ for $y > 0$, so $G(y) = V(y) + (x-y)^2/2t$ is convex on each half-line, with critical points $y_\ell = x - u_\ell t$ (admissible iff $x \le u_\ell t$) and $y_r = x - u_r t$ (iff $x \ge u_r t$), and a kink at $y = 0$ with one-sided derivatives $u_\ell - x/t$ and $u_r - x/t$. If $u_\ell < u_r$ and $u_\ell t < x < u_r t$, only the kink is a local minimiser and $u = x/t$; outside this range the single admissible critical point gives u_ℓ or u_r, in agreement with (28). If $u_\ell > u_r$ and $u_r t \le x \le u_\ell t$, both y_ℓ and y_r are admissible and the kink is a local maximum, with $G(y_\ell) = u_\ell x - u_\ell^2 t/2$ and $G(y_r) = u_r x - u_r^2 t/2$, so $G(y_\ell) - G(y_r) = (u_\ell - u_r)\big(x - \frac{u_\ell+u_r}{2}t\big)$. The tie, hence the jump, is at $x = \frac{u_\ell+u_r}{2}t$, recovering (27) and the Rankine–Hugoniot speed.

8.3.1 Viscous shock profiles

Proposition 8.7 (Viscous shock profiles)**.** *Let $u_\ell \ne u_r$ and $\nu > 0$. Equation* (1) *has a travelling-wave solution $u(x,t) = \phi(x - st)$, with $\phi \in C^2$ bounded, $\phi(\eta) \to u_\ell$ as $\eta \to -\infty$, $\phi(\eta) \to u_r$ as $\eta \to +\infty$ and $\phi' \to 0$ as $\eta \to \pm\infty$, if and only if $u_\ell > u_r$. In that case $s = (u_\ell + u_r)/2$ and, with $\Delta = u_\ell - u_r$, ϕ is unique up to translation $\eta \mapsto \eta + \eta_0$; the translate with $\phi(0) = s$ is*

$$\phi(\eta) = s - \frac{\Delta}{2}\tanh\Big(\frac{\Delta\,\eta}{4\nu}\Big). \tag{36}$$

Moreover $\|\phi - (s - \frac{\Delta}{2}\operatorname{sgn}\eta)\|_{L^1(\mathbb{R})} = 4\ln 2\,\nu$, independent of s and Δ.

Proof. Substituting $u = \phi(x - st)$ gives $-s\phi' + \phi\phi' = \nu\phi''$, and integrating once from $-\infty$ (where $\phi' \to 0$),

$$\nu\phi' = \tfrac{1}{2}\phi^2 - s\phi - K, \qquad K = \tfrac{1}{2}u_\ell^2 - s u_\ell.$$

Letting $\eta \to +\infty$ shows that the right-hand side must also vanish at u_r, so u_ℓ, u_r are the two roots of the quadratic: $s = (u_\ell + u_r)/2$ (the Rankine–Hugoniot speed) and

$$\nu\phi' = \tfrac{1}{2}(\phi - u_\ell)(\phi - u_r). \tag{37}$$

This autonomous first-order ODE cannot be crossed at its equilibria u_ℓ, u_r, so a connecting ϕ lies strictly between them. On $(\min(u_\ell,u_r), \max(u_\ell,u_r))$ the right side of (37) is negative, so ϕ is decreasing. If $u_\ell < u_r$ it would have to increase from u_ℓ to u_r, a contradiction; if $u_\ell > u_r$, (37) is solved by (36); any other solution is a translate, by uniqueness for the ODE. For the L^1 distance, by oddness and $\int_0^\infty (1 - \tanh\zeta)\,d\zeta = \ln 2$,

$$\int_{\mathbb{R}} \tfrac{\Delta}{2}\Big|\tanh\tfrac{\Delta\eta}{4\nu} - \operatorname{sgn}\eta\Big|\,d\eta = \Delta\int_0^\infty \Big(1 - \tanh\tfrac{\Delta\eta}{4\nu}\Big)d\eta = \Delta\cdot\frac{4\nu}{\Delta}\ln 2 = 4\ln 2\,\nu.$$

□

Hence a jump is the limit of viscous travelling waves, of thickness $O(\nu/\Delta)$, exactly when $u_\ell > u_r$. This motivates the Lax condition (23) from vanishing viscosity. That u_A of Section 5, with its upward jump, is not the vanishing-viscosity limit is the content of Theorem 8.4 together with Proposition 6.5.

For Riemann data the Cole–Hopf solution is available in closed form since G is piecewise quadratic (`code/cole_hopf_riemann.py`). For the shock $(1,0)$ its $L^1(-2,2)$ distance to the entropy solution at $t = 1$ equals $2.773\,\nu$ for $\nu \le 10^{-3}$, the layer value $4\ln 2\,\nu$ of Proposition 8.7; for the rarefaction $(0,1)$ the error divided by ν grows by about 4.4 per decade of ν (observed, not derived).

8.4 Godunov's method

8.4.1 The scheme

Let $\mathbb{R}$ be divided into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with $t^n = n\Delta t$ and cell averages $\bar u_j^n = \Delta x^{-1}\int_{x_{j-1/2}}^{x_{j+1/2}} u(x,t^n)\,dx$ approximated by U_j^n. Integrating $u_t + f(u)_x = 0$ over a cell and a time step gives

$$\bar u_j^{n+1} = \bar u_j^n - \frac{\Delta t}{\Delta x}\left(F_{j+1/2}^n - F_{j-1/2}^n\right), \qquad F_{j+1/2}^n = \frac{1}{\Delta t}\int_{t^n}^{t^{n+1}} f\big(u(x_{j+1/2},t)\big)\,dt, \tag{38}$$

for the exact cell averages $\bar{u}_j^n$. The scheme replaces $F_{j+1/2}^n$ by a numerical flux $\widehat{f}_{j+1/2}$ depending on U_j^n, U_{j+1}^n, giving $U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}(\widehat{f}_{j+1/2} - \widehat{f}_{j-1/2})$. Godunov's method [6] takes $\widehat{f}_{j+1/2}$ to be the flux at $x/t = 0$ of the exact Riemann solution with data (U_j^n, U_{j+1}^n) (Theorem 7.1):

$$\widehat{f}_{j+1/2} = f\big(\omega(0;\, U_j^n, U_{j+1}^n)\big), \tag{39}$$

where $\omega(\xi; u_\ell, u_r)$ is the entropy solution with data (u_ℓ, u_r) at $\xi = x/t$. By (27)–(28),

$$\omega(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ \tfrac{u_\ell + u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ \tfrac{u_\ell + u_r}{2} < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{40}$$

the last case corresponding to $\xi = 0$ lying inside the fan. If $u_\ell = -u_r$ the value $\omega(0)$ is ambiguous, but $f(u_\ell) = f(u_r)$, so (39) is unaffected.

8.4.2 Stability and convergence

The scheme is stable and monotone provided the CFL condition

$$\Delta t \le \frac{\Delta x}{\max_j |U_j^n|} \tag{41}$$

holds, which prevents waves from neighbouring interfaces from reaching $x_{j+1/2}$ within one step [10, Ch. 4].

Theorem 8.8 ([4, Theorem 1 and §1])**.** *Let $u_0 \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ with $a \le u_0 \le b$ a.e., and let $\lambda = \Delta t/\Delta x$ be fixed with $\lambda \max_{a \le u \le b} |f'(u)| \le 1$. Let u^Δ be the piecewise constant function equal to U_j^n on $[x_{j-1/2}, x_{j+1/2}) \times [t^n, t^{n+1})$, where U_j^0 is the cell average of u_0 and the U_j^n are the Godunov approximations. Then for every $T > 0$*

$$\lim_{\Delta t \to 0} \sup_{0 \le t \le T} \int_{\mathbb{R}} \big| u^\Delta(x,t) - u(x,t) \big|\, dx = 0,$$

where u is the Kružkov entropy solution [8] with data u_0.

Crandall and Majda establish this for every consistent conservation-form scheme that is monotone on $[a, b]$ with Lipschitz continuous numerical flux, and verify in §1 that Godunov's scheme for $f'' > 0$ belongs to this class; the theorem gives convergence but no rate. For Burgers' equation the Kružkov solution coincides with the Oleinik solution of Theorem 6.6 (standard; see [5, §3.4] and [8]). Since the Riemann data used below are not in $L^1(\mathbb{R})$, we localise: replacing them by 0 outside $[-5, 5]$ keeps u_0 within $[0, 1]$ and, because $|u| \le 1$, leaves the exact solution on $[-2, 2] \times [0, 1]$ unchanged. At $\lambda = 0.45$ the numerical domain of dependence over $t \le 1$ has half-width at most $t/\lambda < 2.23$, so the dependence interval of $[-2, 2]$ lies within $[-4.23, 4.23] \subset [-5, 5]$ and the computed solution is unchanged as well. In the experiments $\max_j |U_j^n| = 1$ throughout, so the adaptive step $\Delta t = 0.45\,\Delta x / \max_j |U_j^n|$ gives the constant $\lambda = 0.45$ required by the theorem (the final step, shortened to land on $t = 1$, has smaller λ).

8.4.3 Numerical results

The scheme was implemented in `code/godunov.py` on $x \in [-2, 2]$ with N cells, taking cell averages of the Riemann data as initial values (for even N the discontinuity lies on a cell interface, while for odd N the cell containing 0 holds a mixed average), outflow boundary conditions, and $\Delta t = 0.45\,\Delta x / \max_j |U_j^n|$, with the last step shortened so that the final time is exactly $t = 1$. The rarefaction data $(u_\ell, u_r) = (0, 1)$ and the shock data $(1, 0)$ were run to $t = 1$, with errors measured in $L^1(-2, 2)$ against the exact solution of Theorem 7.1 integrated cell by cell. As checks, the case-based flux (39) agrees with the closed form $\min_{[u_\ell, u_r]} f$ (if $u_\ell \le u_r$) and $\max_{[u_r, u_\ell]} f$ (otherwise) on 10^5 random state pairs, and total mass equals the initial mass plus the net boundary flux to ten digits. Since the exact solution is constant near $x = \pm 2$ for $t \le 1$, the outflow condition has a negligible effect: repeating $N = 400$ on $[-4, 4]$ with the same Δx changes the computed values on $[-2, 2]$ by at most 3×10^{-14}. Figure 3 shows the case $N = 400$, where the computed shock lies at $x = st = 0.5$ and the fan has slope $1/t$, with no oscillation.

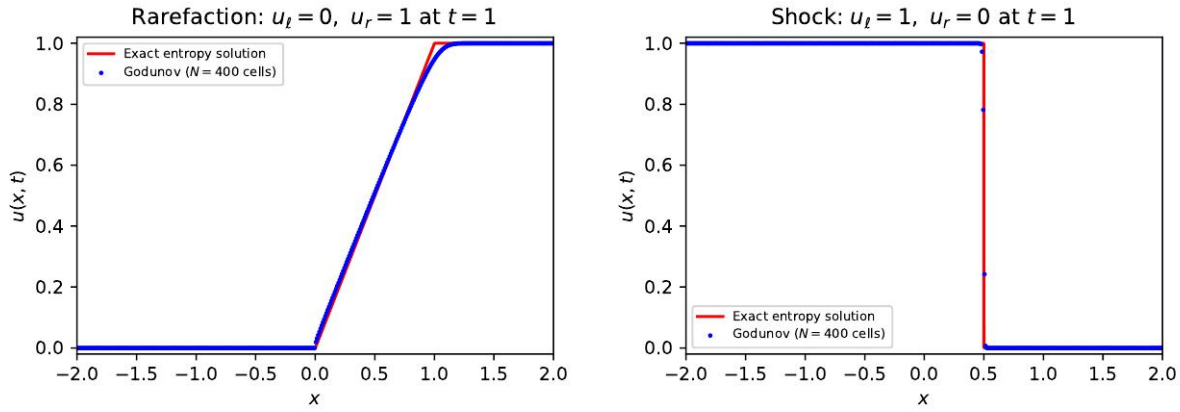


Figure 3: Godunov scheme (dots, $N = 400$) against the exact entropy solution (solid) at $t = 1$. Left: rarefaction data $(0, 1)$. Right: shock data $(1, 0)$.

N	Δx	L^1 (rarefaction)	order	L^1 (shock)	order
200	2.0×10^{-2}	3.05×10^{-2}	—	9.91×10^{-3}	—
400	1.0×10^{-2}	1.82×10^{-2}	0.74	4.98×10^{-3}	0.99
800	5.0×10^{-3}	1.07×10^{-2}	0.77	2.50×10^{-3}	0.99
1600	2.5×10^{-3}	6.14×10^{-3}	0.80	1.24×10^{-3}	1.01
3200	1.25×10^{-3}	3.48×10^{-3}	0.82	6.23×10^{-4}	0.99
6400	6.25×10^{-4}	1.95×10^{-3}	0.84	3.13×10^{-4}	0.99

Table 1: L^1 errors at $t = 1$ and observed orders $\log_2(e_{N/2}/e_N)$. All N are multiples of 8, so the shock lies on a cell interface.

Shock. The shock $x = st = 0.5$ lies on a cell interface iff $8 \mid N$, and for such N the error is $0.50\,\Delta x$ to two digits (0.495–0.501) with observed order ≈ 1 (Table 1). For $N = 97, \dots, 103$ it ranges from $0.55\,\Delta x$ to $0.66\,\Delta x$ ($0.655\,\Delta x$ at $N = 100$, against $0.50\,\Delta x$ at $N = 96, 104$), so orders computed between misaligned and aligned resolutions are unreliable ($N = 100 \to 200$ would give 1.40), which is why the table uses $N = 200 \cdot 2^k$. In all cases the error is proportional to Δx.

Rarefaction. The error decreases monotonically, with observed orders $0.74, 0.77, 0.80, 0.82, 0.84$ that are still increasing, so these experiments do not establish a first-order rate. The error is confined to the fan region (outside $[-0.2, 1.2]$ it is below 3×10^{-4} at $N = 200$ and decays rapidly), but not to the corners: for $N = 200, \dots, 6400$, the error on the fan interior $0.2 < x < 0.8$, which carries about a third of the total, has observed orders $0.75, 0.79, 0.81, 0.83, 0.85$, and the error within 0.1 of the corners has orders $0.65, 0.70, 0.75, 0.79, 0.81$, both rising. A corner effect alone therefore does not account for the sub-first-order rate. One possible explanation is that the numerical layers at the corners, of width $O(\sqrt{\Delta x})$, still overlap the fan at these resolutions; this has not been tested.

8.4.4 Numerical diffusion as vanishing viscosity

For the shock data the scheme is monotone, so by the discrete maximum principle all states remain in $[0, 1]$. All characteristic speeds are then nonnegative, so the Godunov flux is the upwind flux $f(U_j^n)$, and the scheme reads $U_j^{n+1} = U_j^n - \lambda(f(U_j^n) - f(U_{j-1}^n))$, $\lambda = \Delta t/\Delta x$. Taylor expansion gives the modified equation

$$u_t + u\,u_x = \frac{\Delta x}{2}\big(u(1 - \lambda u)\,u_x\big)_x + O(\Delta x^2), \tag{42}$$

a Burgers-type equation in conservative form, with state-dependent viscosity $\nu(u) = \frac{\Delta x}{2}u(1 - \lambda u)$, nonnegative exactly under (41). Its travelling shock $1 \to 0$ of speed $\frac{1}{2}$ satisfies $\nu(u)u_\eta = \frac{1}{2}u(u - 1)$ in $\eta = x - t/2$; in units of Δx,

$$\frac{du}{d\eta} = \frac{u - 1}{1 - \lambda u}, \qquad \eta(u) = \eta_0 + (1 - \lambda)\ln(1 - u) - \lambda u. \tag{43}$$

The profile decays exponentially to 1 as $\eta \to -\infty$ and reaches 0 at a finite η_0, which conservation of mass fixes as $\eta_0 = 1 - \lambda/2$. The L^1 distance from the sharp shock is then

$$\frac{\|u - u_{\text{exact}}\|_{L^1}}{\Delta x} = 2\int_0^{u_0} \frac{u(1-\lambda u)}{1-u}\,du, \qquad (1-\lambda)\ln(1-u_0) - \lambda u_0 = -\Big(1 - \frac{\lambda}{2}\Big). \tag{44}$$

Table 2 compares this with the measured shock error on aligned grids, where λ equals the CFL number since $\max|U| = 1$ (`code/numerical_viscosity.py`).

CFL $= \lambda$	measured ($N = 1600$)	predicted	(pred.−meas.)/meas.
0.10	0.627	0.686	+9.4%
0.20	0.589	0.635	+7.8%
0.30	0.553	0.586	+6.0%
0.45	0.496	0.511	+3.0%
0.60	0.448	0.438	−2.2%
0.80	0.352	0.342	−2.8%
0.95	0.300	0.272	−9.3%

Table 2: Shock L^1 error divided by Δx: measured (grids with $8 \mid N$) against the modified-equation prediction; the last column is computed from the tabulated values.

The prediction is within about 3% at CFL 0.45 (+3.0%) and within 10% throughout, although the value at CFL 0.95 is not yet asymptotic (0.334 at $N = 400$, 0.300 at $N = 1600$). Equating the error with the viscous-shock value $4\ln 2\,\nu_{\text{eff}}$ of Proposition 8.7 ($s = \frac{1}{2}$, $\Delta = 1$) defines an effective viscosity: the measured $0.50\,\Delta x$ gives $\nu_{\text{eff}} \approx 0.180\,\Delta x$, the modified-equation profile gives $0.184\,\Delta x$, and $\nu(\frac{1}{2}) = \frac{\Delta x}{2}\cdot\frac{1}{2}(1-\frac{\lambda}{2})$ gives $0.194\,\Delta x$ at $\lambda = 0.45$. Since $\nu_{\text{eff}} = O(\Delta x) \to 0$, the scheme acts in the shock layer as a vanishing viscosity, in accordance with Theorem 8.4; this is a consistency check rather than a proof, as the numerical shock spans only a few cells and (42) omits the $O(\Delta x^2)$ terms.

8.4.5 A consistent scheme that selects the wrong solution

Consider the flux $\widehat{f}_{j+1/2} = f(U_j)$ if $\frac{1}{2}(U_j + U_{j+1}) > 0$ and $f(U_{j+1})$ otherwise, which is consistent and conservative but not monotone. For the Riemann data $U_j = -1$ ($j \le 0$), $U_j = +1$ ($j > 0$) every numerical flux equals $f(\pm 1) = \frac{1}{2}$, so $U^{n+1} = U^n$ exactly and the scheme returns the stationary upward jump, the analogue for these data of u_A in Section 5 (a weak solution by (18) with speed 0, but not an entropy solution). Godunov's flux at the same interface is $f(0) = 0$, and the resulting scheme produces instead the fan $u = x/t$ of Theorem 7.1. At $t = 1$ the $L^1(-2, 2)$ distance to the entropy solution is 1.000 for the non-monotone scheme for all $N = 200, 400, 800$, since $\int |\operatorname{sgn} x - \operatorname{clip}(x, -1, 1)|\,dx = 1$, while for Godunov's scheme it is 6.1×10^{-2}, 3.6×10^{-2}, 2.1×10^{-2}. By the Lax–Wendroff theorem [10, Ch. 12] limits of consistent conservative schemes are weak solutions, so consistency and conservation alone do not select the entropy solution: monotonicity (Theorem 8.8) does.

9 Conclusion

The weak formulation extends solutions of (2) beyond breakdown but does not determine them: already for Riemann data there are infinitely many weak solutions (Section 5). The Oleinik condition, whose local form at a jump is the Lax condition $u_\ell > u_r$, restores uniqueness (Sections 6–7). It is the physically correct selection because it is the one made by vanishing viscosity: the Cole–Hopf solutions converge to the entropy solution (Theorem 8.4, Remark 8.5), and a viscous travelling wave connecting u_ℓ to u_r exists if and only if $u_\ell > u_r$ (Proposition 8.7), in line with the standard theory [5, §3.4].

The Godunov scheme of Section 8.4 was tested on Riemann data. The shock error is $0.50\,\Delta x$ on grids aligned with the shock and 0.55–$0.66\,\Delta x$ on the misaligned grids tested ($N = 97, \dots, 103$); the rarefaction error has observed order ≈ 0.8, still increasing. The modified equation (42) predicts the shock error to within about 3% at CFL 0.45 and within about 10% for CFL numbers in $[0.1, 0.95]$, corresponding to an effective viscosity $\nu_{\text{eff}} \approx 0.18\,\Delta x$. This is a heuristic consistency check; the rigorous result is Theorem 8.8 [4].

Theorems 6.6, 8.4 and 8.8 are taken from the literature, and the long-time N-wave asymptotics for compactly supported data [9], [5, §3.4.5] lie outside the scope of this paper.

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